MATLAB Processing Pipeline for Emotiv EPOC X EEG output + Local Power Spectrum Viewer
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Project Description
I need a complete, local-only MATLAB (EEGLAB preferred) solution for Emotiv EPOC X EEG data.

Goal
1. Capture EEG data from individuals using the Emotiv EPOC X (14 channels), as an EDF or BDF file. (Note: An EDF (or BDF) file is a standardized binary format that stores multi-channel time-series physiological signals (such as EEG) together with header metadata including sampling rate, channel labels, physical units, and recording timestamps.)
2. Automatically process the data (noise filtering + artifact removal).
3. Save the cleaned data locally.
4. Generate and display **power spectrum plots** of the cleaned recordings.
5. Allow viewing of spectral power across the 0–20 Hz band for each channel, and comparing multiple recordings over time.
6. Everything must run and be stored **locally only** (no cloud) for data privacy and safety.

Required Preprocessing Pipeline
1. Inspect and reject bad segments
2. High-pass filter (0.5–1 Hz)
3. Low-pass filter (40–45 Hz)
4. Optional notch filter (50/60 Hz)
5. Re-reference (e.g. Common Average Reference)
6. Artifact removal using ICA (+ ICLabel) and/or ASR
7. Bad channel interpolation
8. Epoch rejection

Required Output
After processing, the system must generate clear **power spectrum (PSD) plots** of the cleaned EEG data. This data should be generated via Fourier Transform.

I will attach a picture showing the desired display style (how the power spectrum should look). The plots should support viewing single recordings and comparing multiple recordings over time.

Deliverables
- Clean, well-commented MATLAB/EEGLAB script(s) for the full automatic pipeline
- Batch processing capability for multiple recordings
- Local storage of raw + cleaned data in a clear folder structure
- Local viewer/script that displays power spectrum plots and allows comparison of multiple recordings. Ideally display 3 recordings at once: the first recording, the second last recording, and the last recording.
- Brief README with clear instructions

Technical Details & Resources
- Device: Emotiv EPOC X (14 channels: AF3, F7, F3, FC5, T7, P7, O1, O2, P8, T8, FC6, F4, F8, AF4)
- Preferred tools: MATLAB + EEGLAB (with ICA, ICLabel, and ASR)

Official Technical Specifications / User Manual:
https://emotiv.gitbook.io/epoc-x-user-manual/introduction/technical-specifications

Channel location file (.ced):
https://raw.githubusercontent.com/huytungst/EEGEmotions-27/main/emotivX_channels_location.ced

Sample Emotiv EPOC X / EPOC data files (publicly available):
- Harvard Dataverse (EPOC X recordings): https://doi.org/10.7910/DVN/JMH4PD
- Zenodo (14-channel Emotiv EPOC recordings): https://zenodo.org/records/1183360
- EmoX dataset (recorded with Emotiv EPOC X): https://ieee-dataport.org/documents/emox

Example of ICA artifact removal (for reference):
https://spkit.github.io/auto_examples/electroencephalogram/plot_sp_eeg_artifact_removal_ICA.html

Skills Required
- Strong MATLAB + EEGLAB experience
- EEG preprocessing (filtering, ICA, artifact removal)
- Experience with Emotiv EPOC / EPOC X data is a strong plus

Please confirm in your bid:
1. That you can deliver a fully local solution
2. You can automatically capture the EDF output, run it through the processing in the background, and automatically display a power spectrum plot.
3. You have experience with cleaning of EEG data.
4. You can create the power spectrum plot layout desired
5. Your expected turn around time range.

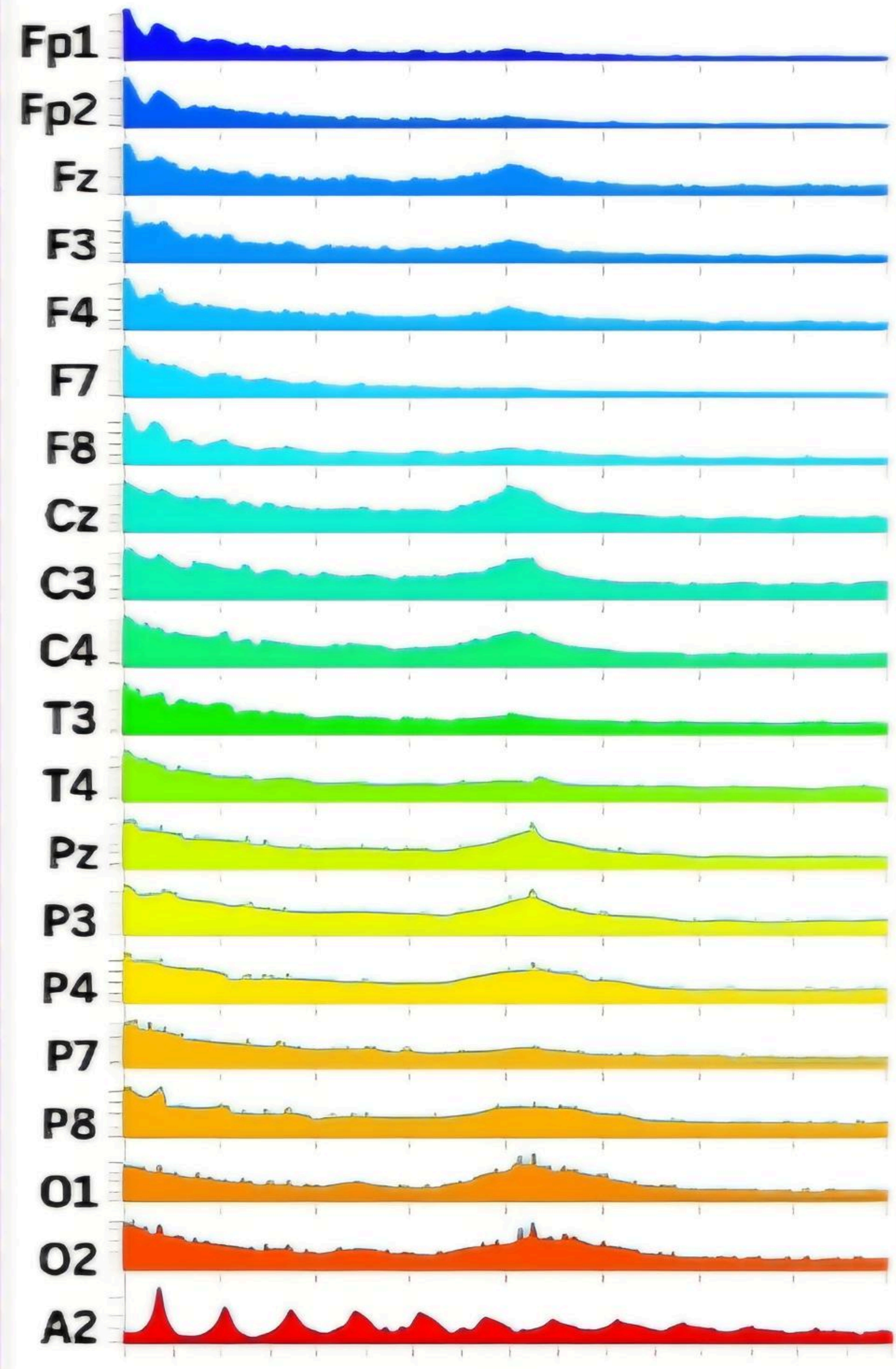
Fp1
Fp2
Fz
F3
F4
F7
F8
Cz
C3
C4
T3
T4
Pz
P3
P4
P7
P8
O1
O2
A2

Figure: desired power spectrum plot